

ROXAS, Philippines

WMO: 985380

Lat: 11.598N

Lon: 122.752E

Elev: 10

StdP: 14.69

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	74.9	75.6	69.5	108.4	78.5	70.7	113.3	78.6	22.6	77.2	20.4	78.6	8.3	40	0.393

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	10.9	91.6	82.1	90.7	81.8	90.0	81.5	82.8	90.0	82.3	89.5	81.9	88.8	8.2	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.6	159.1	88.8	80.1	156.1	88.1	79.8	154.5	87.8	46.6	90.6	46.1	90.0	45.7	89.5	86.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.5	15.1	13.4	72.0	95.5	3.3	3.4	69.7	97.9	67.7	99.9	65.9	101.9	63.5	104.4		
(5)			67.5	84.2	3.2	1.0	65.2	85.0	63.3	85.6	61.5	86.1	59.2	86.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	82.5	80.3	80.7	82.0	83.7	84.4	83.4	82.7	83.0	82.7	82.6	81.5	(6)	
(7)		DBStd	2.02	1.66	1.49	1.69	1.64	1.76	1.78	1.70	1.53	1.65	1.61	1.66	1.79	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD50	11856	938	860	993	1011	1067	1003	1014	1022	980	1012	978	978	(10)
(11)		CDD65	6381	473	440	528	561	602	553	549	557	530	547	528	513	(11)
(12)		CDH74	71808	4472	4369	5817	6930	7671	6547	6173	6253	5852	6225	6043	5456	(12)
(13)		CDH80	24092	798	948	1759	2777	3374	2550	2153	2199	1992	2144	1984	1414	(13)
(14)	Wind	WSAvg	6.6	9.1	8.1	7.7	6.8	5.4	4.8	4.7	4.9	5.8	7.9	9.4	(14)	
(15)	Precipitation	PrecAvg	91.1	4.7	3.5	3.5	2.6	6.2	10.5	11.1	7.2	8.5	12.1	11.0	10.0	(15)
(16)		PrecMax	160.9	8.3	10.5	11.7	13.3	16.1	30.9	20.5	17.5	19.2	20.7	20.6	21.1	(16)
(17)		PrecMin	39.4	2.0	1.1	0.7	0.3	0.9	2.7	4.1	2.1	2.2	3.7	4.1	3.7	(17)
(18)		PrecStd	24.1	1.8	2.6	3.0	3.4	4.2	5.6	4.4	3.8	4.7	4.4	4.6	4.8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	87.3	87.8	89.6	91.2	93.2	92.2	91.6	91.6	91.5	90.7	90.0	89.1	(19)
(20)			MCWB	80.1	80.3	81.0	82.0	82.4	82.3	82.0	81.8	81.8	81.9	81.8	81.4	(20)
(21)		2%	DB	86.1	86.4	88.3	90.0	91.6	90.9	90.0	90.3	90.0	89.5	88.9	87.7	(21)
(22)			MCWB	79.6	79.5	80.3	81.5	82.3	82.1	81.4	81.4	81.3	81.4	81.3	80.8	(22)
(23)		5%	DB	84.9	85.5	87.3	89.2	90.6	90.0	89.1	89.3	88.9	88.5	88.0	86.6	(23)
(24)			MCWB	79.0	79.1	79.9	81.1	81.9	81.7	81.0	81.0	80.9	80.9	80.8	80.4	(24)
(25)		10%	DB	83.8	84.6	86.4	88.4	89.6	88.9	87.9	88.2	87.8	87.5	86.9	85.5	(25)
(26)			MCWB	78.4	78.4	79.4	80.6	81.5	81.2	80.6	80.6	80.5	80.4	80.4	79.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.9	81.1	81.9	82.8	83.5	83.3	82.7	82.7	82.6	82.6	82.6	81.9	(27)
(28)			MCD B	85.9	86.3	87.9	89.7	91.1	90.7	90.0	90.0	89.8	89.3	88.8	87.8	(28)
(29)		2%	WB	80.1	80.3	81.2	82.1	82.7	82.5	81.8	81.8	81.8	81.8	81.8	81.2	(29)
(30)			MCD B	85.1	85.5	87.2	89.0	90.4	89.9	88.9	89.1	88.8	88.3	88.0	86.8	(30)
(31)		5%	WB	79.4	79.7	80.6	81.6	82.2	81.9	81.3	81.3	81.2	81.3	81.2	80.6	(31)
(32)			MCD B	84.1	84.7	86.6	88.4	89.7	88.9	88.2	88.3	88.0	87.6	87.1	86.1	(32)
(33)		10%	WB	78.7	78.9	79.9	80.9	81.7	81.4	80.7	80.7	80.7	80.7	80.6	79.9	(33)
(34)			MCD B	83.2	83.8	85.6	87.6	88.9	88.1	87.3	87.5	87.3	86.9	86.4	85.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.6	7.5	8.6	9.7	10.9	10.9	10.5	10.7	10.5	9.8	8.7	7.4	(35)
(36)			MCD BR	8.2	8.8	9.8	10.5	11.9	12.2	12.0	12.1	12.0	11.1	10.1	9.1	(36)
(37)			MCWBR	4.7	4.8	5.1	5.2	5.7	6.0	5.8	5.8	5.8	5.6	5.3	5.1	(37)
(38)		5% WB	MCD BR	7.9	8.4	9.4	10.4	11.4	11.8	11.5	11.7	11.6	10.8	9.8	8.8	(38)
(39)			MCWBR	4.8	5.0	5.3	5.5	5.8	6.0	5.9	6.0	6.0	5.8	5.4	5.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.440	0.428	0.431	0.436	0.436	0.432	0.428	0.442	0.436	0.453	0.443	0.446	(40)	
(41)		taud	2.391	2.417	2.416	2.402	2.425	2.447	2.459	2.408	2.421	2.365	2.395	2.373	(41)	
(42)		Ebn at Noon	274	282	282	278	273	271	273	272	277	271	271	269	(42)	
(43)		Edh at Noon	38	38	39	39	37	36	36	38	38	39	37	38	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1288	1574	1793	1964	1792	1624	1499	1529	1530	1477	1431	1249	(44)	
(45)		RadStd	143	163	182	107	161	116	126	100	146	99	115	94	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.67	+0.90	N/A	+0.83	+0.32	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	+0.29	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page